
Research Statement

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I am a researcher working in macro-finance and banking. I study how banks and firms interact in the market for credit: what determines of loan pricing, how credit is allocated across firms, and how monetary policy affects equilibrium outcomes. My investigation starts from an empirical fact: observationally equivalent firms in terms of credit quality do not pay the same rates for similar loans; my research shows that this is true when firms borrow from different banks, from a bank and a non-bank, or even when they borrow from the same bank. Understanding these rate differentials matters because they have implications for regulation and monetary-policy transmission, as well as for firms' investment and growth.

The big puzzle that organizes my work is to understand how much of the price a firm pays reflects what its lender knows about the true credit quality of the firm, and how much reflects how little competition that bank faces. The question is hard because the two forces are not independent. A bank prices on what it learns only to the extent that rivals cannot undercut it, and it pays to learn only when the rents from screening survive competition. Information and competition therefore jointly determine the rate a firm is quoted, which firms are offered credit at all, how the surplus from a lending relationship is divided, and how monetary policy passes through to firms. Studying each of them in isolation lead to misinterpreting all four outcomes. In my research output, I address this question from several perspectives, relying on administrative French micro-data on lending, and equilibrium models of loan pricing under search frictions and asymmetric information.

Job market paper. In my job market paper, I test whether there is evidence of banks knowing more than the econometrician when looking at rate differentials. This information is, by construction, private to the bank; I exploit the fact that, if banks' expectations about firms' credit quality are correct, they should be reflected in realized outcomes. I use public rating data by Banque de France, a sufficient statistic for creditworthiness, to track the evolution of firms' default probability after loan issuance. Using about one million French loans matched to Banque de France ratings, I find that rates contain information on future revisions of default probability: a firm whose probability of default increases by 1 percentage point after one year paid, on average, 0.04 percentage points more at issuance; the premium doubles for young and small firms, is larger at long maturities, and peaks in 2009-2012. Comparing fixed- and variable-rate borrowers when reference rates rose rules out the reverse channel: a more expensive loan does not lead to the firm's rating deterioration. However, information cannot be the whole story: I build an equilibrium model in which banks post rates under search frictions and screen on an imperfect public signal, and I show that search frictions alone can generate rate dispersion among identical firms and can induce self-selection through pricing. I use the data to discipline the model, calibrating it separately to each of the eight rating classes to assess the relative importance of these forces. I find that in safer markets, banks are informed and separate borrowers; in riskier ones, they are uninformed and pool borrowers. Information determines who is served and on what terms: better signals allow lenders to separate safe from risky firms, reducing cross-subsidization and preventing safe firms from being priced out of the market.

Expanding the analysis to the euro area. The job market paper is constrained by data limitations that I aim to overcome by accessing the euro area central bank loan level registry,

AnaCredit. My current measure of bank information relies on public ratings, and it has a clear bias: I observe a revision only for firms whose rating is actually reviewed, not for every firm the bank expects to deteriorate, so the approach *understates* how much rates reflect what banks know at issuance. Observing banks' own internal ratings would close that gap. Moreover, I could raise a question worth asking in its own right: are internal ratings accurate in measuring expected default risk, or do they respond to other banks' needs related to capital requirements and regulation? I plan to test how precisely internal ratings assigned at issuance predict realized default. AnaCredit provides the perfect setting to explore this question: it records internal ratings alongside loan terms, the full bank–firm exposure history and default outcomes, none of which survey data provide. In addition, the registry covers banks and firms across the euro area, allowing me to study cross-country differences in monetary-policy pass-through and assess their implications for policy. My model provides a framework to interpret these differences through country-specific parameters governing signal precision and the degree of competition.

A model of information acquisition. I am currently expanding the theoretical framework of my job market paper to understand endogenous information acquisition dynamics under asymmetric information and imperfect competition. While the current model captures the main empirical patterns well, it assumes that banks observe the same signal about a firm, so the asymmetry runs between banks and firms, and between the econometrician and banks. The extension allows banks to conduct independent screening, so that their signals need not be perfectly correlated and may differ in precision. This allows me to characterize the equilibrium choice of screening intensity and study how banks' incentives to acquire information depend on the competitive environment and on the information acquired by other banks. Preliminary results suggest that dispersion persists across the whole range of feasible rates. The main difference is that the signal segments the distribution of rate offers within each bank: the supports of the rates associated with different signal realizations do not overlap, whereas they do in the current model. Also, as long as search frictions remain, the same empirical prediction holds: information acquisition alone cannot fully account for the observed price dispersion, and interactions between information and search remain essential.

Banks and peer-to-peer lenders lending practices and monetary policy response. My current work also expands the analysis to non-bank lenders, and their interaction with the banking system. I am working with Mattia Girotti (Université Paris Dauphine) and Andrea Polo (LUISS) to compare how banks and registered peer-to-peer lending platforms pass monetary policy shocks through to firms, using French loan-level data that cover both, and web-scraped data from platforms' websites. The rate dispersion fact remains: we observe similar firms, sometimes the same firm, borrowing from banks and platforms at several percentage points difference. Platforms are a natural comparison group to banks for investigating how private lender information affects pricing: they collect, publish, and process borrower information differently, and relationship lending dynamics are much less strong in this setting. The project is in progress, but I can already document very different lending behaviours and monetary policy responses. First, within platforms, there is virtually no dispersion among firms with the same credit score, assigned by the platform itself based on internal algorithms. Second, in response to a monetary shock, the two lenders respond very differently. While banks contemporaneously raise rates and cut quantity, the main channel through which platforms respond is quantity. Moreover, by observing data on investor type from the loan registry complemented with web-scraped data, we document heterogeneity in the response by investor type: the tightening is driven by banks themselves and other institutional investors lending via the platform, while individual investors do not respond significantly.

Two challenges remain open. First, firms borrowing from peer-to-peer platforms tend to be relatively small, so the main challenge is to recover additional data. We are tackling this by expanding the data collection using administrative data from INSEE on firms' demographics

that are absent from the credit registries. A second, and related, question is to understand whether firms borrowing from platforms are systematically different from those borrowing from banks, and whether the two groups are substitutable. We are addressing this by matching the two samples on observables, and also by collecting and constructing firm-level default/distress events, such as liquidation or judicial restructuring, using the French administrative source BODACC.

Lender heterogeneity in the pricing of bundled credit. Further exploring the lending behaviour of banks, I am looking at an understudied dimension of bank pricing: bundling. I ask how banks price bundled products. While empirical evidence is lacking, the literature would predict discounts on bundled products at the beginning of the relationship, to attract borrowers and lock them in, and/or because bundling lowers the cost of monitoring them - for example, when loans are combined with deposits or overdrafts. I document several facts. Using the same French administrative loan-level survey over the zero lower bound period, I can observe whether multiple products are issued contemporaneously to the same firm. First, I document that bundle pricing varies across lenders: the average gap is -0.05 percentage points, close to zero, but it hides a spread of 1.69 p.p.: of 99 lenders, 19 price bundles at a significant premium, 15 at a significant discount, and the rest show no significant gap. Second, I flag first-time borrowers as those for whom there is no record of lending with the bank in the previous quarters, and I find that the bundle gap does not move with the relationship. I decompose how much of the observed heterogeneity comes from borrower composition and product mix, which respectively explain 0.6 and 4.9 percent of lenders' heterogeneity, while the rest points to lender-specific strategies.

More work needs to be done on the empirical and theoretical sides. I plan to test two directions. First, drawing from my job market paper methodology, testing self-selection via menus: firms paying a premium on a bundle may be systematically riskier and may self-select into more expensive bundles, vice versa for safer ones. I will test whether default outcomes support this hypothesis; if not, this would suggest that the observed pricing heterogeneity is not driven by risk-based self-selection, pointing instead to mechanisms such as price discrimination or borrowers' myopia. These findings can motivate new theories of bank bundling strategies. Second, I will explore long-term trends, exploiting the full history of the survey, to see how bundling strategies evolve over time and over different rate regimes.

Taken together, my research studies how information and competition shape the pricing and allocation of credit. Across banks and non-bank lenders, I aim to understand how lenders' private information, market power, and strategic choices generate differences in loan pricing, credit access, and monetary-policy transmission. Going forward, I aim to combine richer loan-level data with equilibrium models of information acquisition and competition. This agenda will allow me to study how changes in information sharing, market structure, and lender organization affect credit markets, and ultimately to distinguish when credit-market dispersion reflects valuable information and when it reflects market power.